AGI ALPHA FLYWHEEL — When Capability Creates Opportunity

Transformational Intelligence, Experience-Trained Specialists and the Expanding Frontier of Profitable Work

Vincent Boucher — President of MONTREAL.AI

EVIDENCE COMPANION · 27 OCTOBER 2026

How to read the evidence

Observed execution is not observed ecology or demand. Sourced facts remain limited by their source and date. Assumptions drive conditional simulations. Unknown quantities remain unknown. This edition claims no calibrated biological, demand or operating-cost model.

1 Identity, chronology and scope

All paths refer to `studies/agi-alpha-purpose-preserving-opportunity-v1/`. The inspected repository base is `47c17f6aca6b1bb8779dd3e47d5ba5f8af3329f4`. Existing orchestration, lifecycle, access and accounting interfaces are reused; prior papers, corrections and active teams are preserved.

Record	Meaning
`DISCOVERY_PROTOCOL.json`	27 October 2026, 16:30:50 UTC. Six serious candidates; 16 initial primary documents plus a targeted three-document extension. Explicit additional source categories.
`METHOD_FREEZE.json`	16:37:51 UTC. Compiler, catalog, independent checker and acceptance protocol fixed before fresh cases.
`CONFIRMATION_FREEZE.json`	Under `evaluation/`, 16:40:33 UTC. Two fresh packets and expected reports committed after the method freeze.
`WORKFLOW_PROTOCOL.json`	16:41:49 UTC. Three same-model conditions, two rounds, 480 seconds per round; no external feedback before B.
`workflow_evidence/`	Original tasks, catalog, reports, receipts, retained scripts and exact task assignments. Scripts archived unchanged with `.py.txt` suffix.
`REVIEW.md`	Post-trial internal review, five diagnostic failures, replay repair and interpretation limits.

Compiler SHA-256:

`f6fac1a211c884735bbdaab451e74d73e4877ad12556ffa22e3a02bbb63c0590`

Catalog SHA-256:

`6a22183309aae61954b4dcf27e643150c034bd6bdf86362cf1ce9b55ab541010`

The catalog contains 21 selected factual taxon annotations, principally from the Xerces Upper Midwest guide (2015, pp. 12, 13, 15), with a separately sourced pollen-only exception. Documented mix contexts are not universal habitat classifications. Most nectar fields are unknown. Lots, prices, quantities, dates, provenance regions and charter thresholds are constructed. The original Illinois workbook could not be downloaded (HTTP403); no workbook or original mixture reanalysis is claimed.

2 The inspectable lineage

Link	Direct evidence	What it does not establish
Need and selection	`DISCOVERY.md`; six candidate records and primary sources	Current buyer demand or a unique new seed solution
Restriction witness	`results/information_loss_witness.json`; identical purchase projection, distinct charters	Autonomous recovery of missing intent
Development delivery	`results/development_independent_check.json`; correct negative audit with two blocked rows	Customer acceptance or acquired seed
Frozen confirmation	`results/fresh_confirmation.json`; 24/24 correct	General contract compliance or ecological validity
Actual recipient reuse	E+ receipts bind unchanged compiler bytes on dry and wet/host work	An independently operating customer organization
Complete accounting	Actual zero collections and unknown complete cost; explicit conditional ledgers	Actual profit, savings or earned surplus
Selective retention	Research package retained; speculative economic admission stopped	Profitable compounding or specialist ASI

3 Acceptance and failure evidence

Fresh confirmation: 24/24 correct. Eight cases are positive documentary screens, thirteen valid negative reports and three invalid inputs. The second scenario changes habitat context, host and provenance. Correct rejection is accepted output under the audit contract. These are not 24 accepted purchases, restored sites or customer projects.

The checker was authored by a separate same-session agent without inspecting the compiler. Source and method freezes precede fresh case construction. Protected material was excluded from workflow actors and capability delivery. Procedural isolation in a shared filesystem is not a cryptographic confidentiality boundary. Protected answers are now disclosed for reproducibility, so subsequent reuse of them is known-bank checking.

Five broader defects are observed

`review/diagnostics.json` contains the full mutated inputs, expected reports and actual reports or exception. Omitted required price defaults to zero; unsupported host strings are accepted; missing source traceability can pass; empty lot identifiers are not canonicalized; and a missing nectar key can raise an uncaught exception. These selected post-trial cases are separate from the original score. The frozen compiler is not generally production qualified.

Meaning limits. The charter and structured mapping are supplied by researchers. Preserving its text does not prove complete semantic extraction. All proposed rows serve the single declared resource mode. A mixed nectar/pollen/host design needs a new contract. Unknown nectar is not absent nectar; guide flowering intervals do not establish nectar output, viable seed, survival or restoration outcomes. Source screening never sets customer acceptance or production authority to true.

4 Workflow observations and fair comparisons

Arm	A seconds	B seconds	Correct	Existing capability permitted
H	—	—	Not run	Equivalent human service; estimated scope/rate only
U	87.000	29.000	24/24	Full facts, coding, checks and own retained script
E	107.797	44.361	24/24	Same resources plus explicit expert procedure
E+	82.657	41.686	24/24	Same expert procedure plus frozen compiler

All model arms had a 480-second cap per packet, identical evidence and tools, and no oracle feedback between A and B. Every arm recovered a truncated initial display by rereading the facts. Underlying execution operations were U 5/2, E 5/3 and E+ 3/2. No post-grading correction enters the primary score. Raw receipts record actual checks and retries; their detail is not normalized into invented labor time.

There is one actor per condition. Totals are U 116.000 seconds, E 152.158273 and E+ 124.343284. E+ is 27.814989 seconds faster than E and 8.343284 slower than U across these executions. The accepted-count E–U and E+–E differences are zero. No population effect, provider benchmark, statistical confidence interval or matched-dollar advantage is established. Provider response IDs, token bills, genuine customer effort and pre-command dispatch time were unavailable.

`results/workflow_assay.json` aggregates grades without altering outputs. The 72 reports are three conditions on the same 24 tasks, not 72 independent tasks. U and E also retain their own newly written scripts. The ordinary-library control uses identical compiler bytes and reproduces all results; no graph-specific benefit is assigned. Actual artifact inheritance is demonstrated, and incremental value against competent alternatives remains unproven.

5 Research artifact and distribution boundary

Version 1.0.0 is a standard-library Python instrument with a catalog, manifest, provenance, dependencies and scope. Its implementation is Apache-2.0; source facts retain attribution. No proprietary weights were modified and no Gym campaign was run. There is no post-Gym version to compare. The capability bundle excludes protected answers, evaluation packets and diagnostic answer material. Research retention is distinct from economic admission and production activation.

6 Economics: labels and accounting identities

Quantity	Evidence status
Sales / customer collections	Observed study state: zero / USD zero
Complete actual research cost	Unknown; no complete provider/engineering bill exposed
Actual profit / available earned funds	Unknown; cannot be inferred from zero sales or unobserved costs
H reference	Assumed three team hours at USD 100/hour; not a quote
U/E/E+ equal labor	Uncalibrated mechanical null control; not required human review or realistic automated cost
Price, demand and probabilities	Assumed; no buyer validation or calibration
224 economic evaluations	Simulated conditional outcomes; mutually exclusive worlds are not customers

Read `economics/NULL_CONTROL_NOTE.md` before any commercial interpretation. The frozen control explores four configurations, six price/deposit combinations and eight worlds (192), plus four technical/participation prerequisite cases across eight worlds (32). It preserves decline, correction, common failure, default,

withdrawal, partial delivery and lateness.

A/B/C. Complete paid delivery, including necessary learning and required shared construction, belongs in A. Additional assessment, admission, sharing, serving and upkeep beyond A belong in B. Optional additional Gym/R&D belongs in C. An already included learning subset is not charged again. Actual unpaid study work is research, not customer-funded A. Paid monetary labor and the same displaced-production opportunity cost must not both be charged.

Cash versus contribution. Refundable deposits remain liabilities and restricted reserves until accepted delivery. Capital is not revenue. Accepted fees, receivables, cash collections, credit losses and refunds are separate. A receivable writeoff is not a second cash payment; a deposit refund is not a second expense. Funds available equal cash less restrictions/reserves. Every cash prefix matters, not only ending profit.

At the illustrative USD 400 price and half deposit, expected complete contribution is H USD 33.05, U/E USD 28.6625 and E+ USD -73.1375. Required funds are USD 350, 355 and 457 respectively. E+ exceeds assumed USD 400 starting funds by USD 57. The numbers describe the null control only. The original selector ranks contribution before liquidity and is not a general joint optimizer; the separate policy addendum preserves that limitation and tests the correction without changing old results.

For combined hypothetical physical construction, the model assumes sufficient evidence and compatible conditional participants. It produces USD -742 expected contribution and USD 7,320 required capital versus USD 4,000. Actual evidence and participation remain missing. The conventional expert tie is logical by construction, not independent field measurement. Native integration's separate fixture amounts are interface tests and must never be added to these results.

Retention boundary. `results/retention_boundary.json` stores 56 assumed sensitivities. With fixed incremental spend USD 100 and serving USD 2/reuse, $N(\delta - 2) - 100 > 0$. At 25 reuses, gross incremental benefit must exceed USD 6/reuse. No observed dollar benefit or demand supports those input values. No fixed reinvestment fraction or mandatory training is inferred.

Prospective policy correction. The original selector ranked contribution before checking liquidity. A separate source/contract freeze preceded six fresh diagnostic sets containing twenty hypothetical alternatives. All six complete expected rankings match. The corrected selector chooses a funded service over a higher-contribution illiquid plan, permits affordable worthwhile retention, stops on unknown readiness or negative contribution, and selects a better conventional alternative when supplied. It checks every declared cash prefix, including zero-weight stress worlds. `POLICY_ADDENDUM.md` preserves the limits: declared attestors are not authenticated, assumed demand is not discovered, and no actual profit or authority is established. Original economic results are unchanged.

7 Native interfaces and authority

`results/native_integration.json` records 18/18 checks against actual imported Journal, TrustRegistry, ResourceAccess, CapsuleManifest and CostLedger interfaces. Sixteen imported native file identities are bound in the inspection record. Exact resource/mission binding, expiry, replay, revocation, wrong scope, separate current mission approval, duplicate charges and unknown costs are exercised.

The supported invocation inspects source bytes. Temporary local keys and fictional ENS associations do not authenticate live agents. The unsupported ecological capsule is refused; current Jobs/settlement roles and production trust roots remain unchanged. A passing signature proves an identity relation under the fixture, not scientific truth, external customer acceptance or permission. Stable qualification may be reused only when its identity and scope match; fresh meaning and authority remain separate.

8 Reproduction and verification

Run `reproduce.py` with a fresh output directory. It checks all linked method and confirmation hashes and lengths, exact packet membership, catalog identity, expected answers, workflow archive hashes and E+ implementation identity before executing the compiler and regrading saved reports. It copies the artifact to an ordinary library for the equal-content comparison and replays conditional economics. The optional `--native` route requires the pinned parent repository and dependencies.

`review/replay_validation.json` records seven rejected corruptions, relocated standalone success, in-repository native success and byte-identical economics. It also documents that the first replay wrapper lacked some hash checks; the corrected wrapper is outside the frozen core. A repaired replay cannot make the core's five defects disappear. Saved-output regrading is not rerunning the model actors or remeasuring their timings.

Local targeted verification covers the instrument, independent oracle, replay, native adapter and economics. Seven native unit tests and nine accounting tests were run. Lint/type outcomes and final document/browser checks are recorded in the package verification record. The repository-wide environment is not fully qualified: required dependency check passed, but full setup encountered inherited conflicts; Docker, Compose and primary pre-commit were missing, and Node 24.22.1 differs from the required 22.17.1. Full hosted release qualification remains pending.

9 Exact foundation identities

The papers were read through local exact counterparts at the frozen repository base. A failed retrieval of a supplied Pages URL does not become a claim that served bytes were downloaded and matched. Full TeX and PDF identities are in `FOUNDATIONS_AND_MODELS.md`.

Foundation PDF	SHA-256
Transformational Intelligence	`d7170e8217e3bf6aa1b84868c1c76c5994c6995c5297ed092d7e72a6430c664f`
Invent the Economy	`61e92a47b32cbca12b705094b982dddaca03b3dfe2b7f657bbf30eb8d56bba79`
SUCCESSOR Ω Harness	`640cba6e0371407577f708856efa4e4d3b685ed7349cecf48491f1660563fda0`

Official Astra, Terra, current model-guide and pricing pages were inspected on 27 October 2026. API identifiers and published prices inform proposed configurations only. Account access and limits, actual response IDs and the session bill are unverified. ChatGPT Pro is not an API model slug or entitlement. No proprietary model weight training, upgrade comparison or live computer-service sale is claimed.

10 Publication, rights and provenance

The stable coordination location is https://github.com/MontrealAI/AGI-Alpha-Agent-v0/issues/4850. Scope comments 6447449795 and 6447638743 precede the final handoff. Existing release, merge and Pages procedures remain in force. A draft research PR or immutable source/PDF link is not a production release or deployed website. Exact public handoff information belongs in `PUBLICATION.md` once verified; no placeholder URL counts as publication.

AI assistance covered source research, implementation, checking, analysis, translation and drafting under the stated author direction. Different same-session roles improve error detection but do not provide external peer review. The catalog contains attributed selected facts; complete source texts and photographs are not redistributed. The reproducibility distribution discloses protected answers only after testing; the capability distribution excludes them. No secrets, real account credentials or production signing keys are included.

Qualification record

Established: bounded documentary correctness, actual artifact transfer, equally correct competent baselines, explicit failures and executable conditional accounting.
Unestablished: independent beneficiary acceptance, a newly feasible ecological project, complete-cost advantage, verified demand, profitable retention, live network admission and commercial compounding.